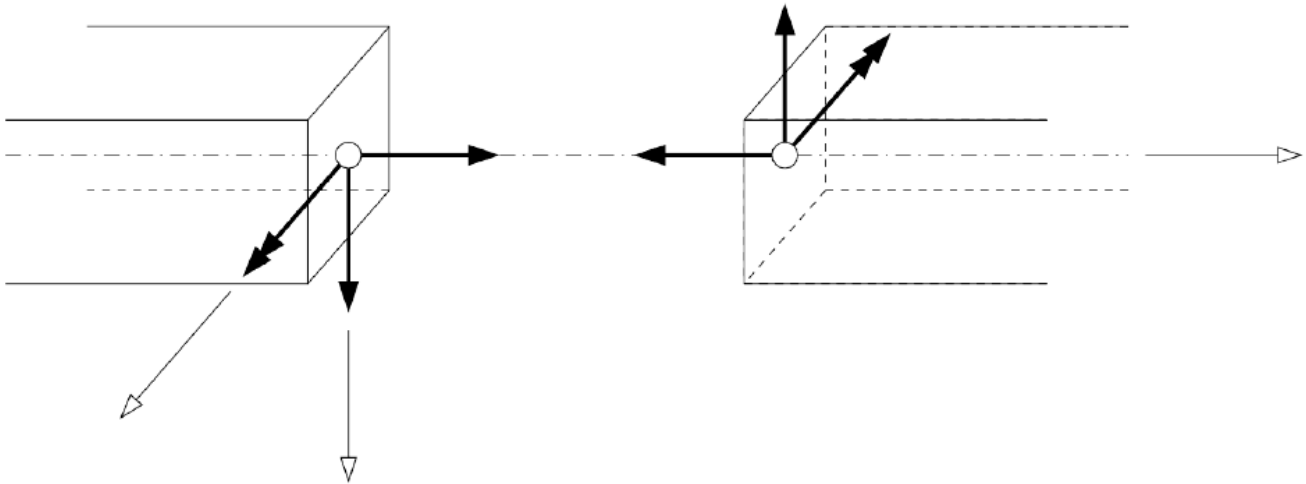
# Engineering Mechanics I

## Lecture



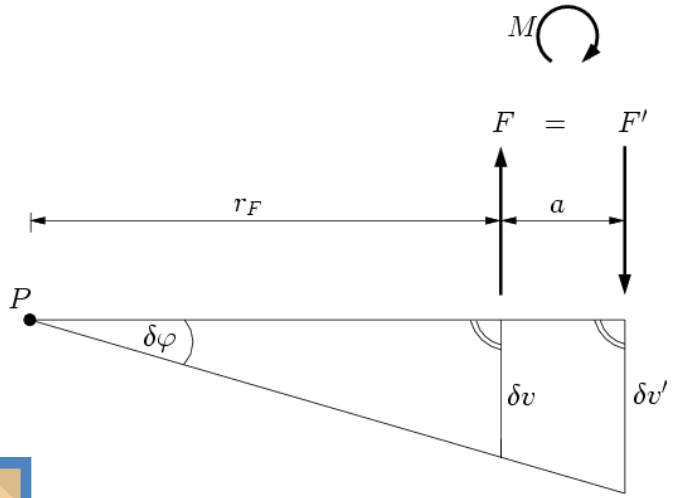
Autonomous Aerial Systems

# 10.3.2. Virtual Work

- Virtual work of a moment over a virtual rotation:

$$\delta W = \vec{M} \cdot \delta\vec{\varphi}$$

Verification via force pair M = aF:

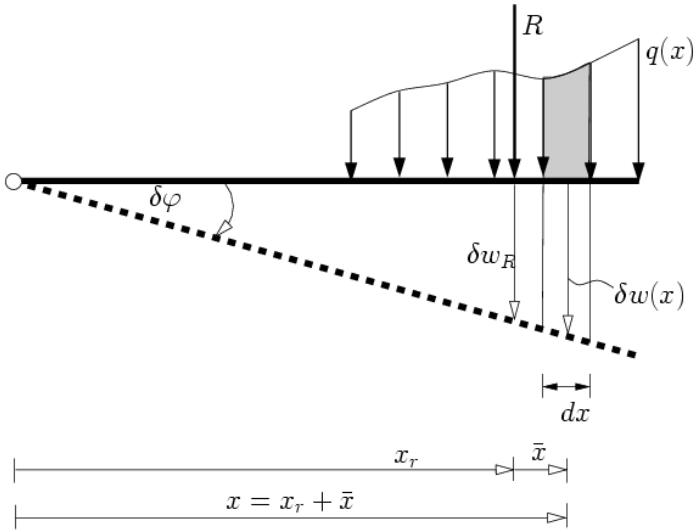



$$\delta v = \delta\varphi \cdot r_F$$
$$\delta v' = \delta\varphi \cdot r_{F'} = \delta\varphi \cdot (r_F + a)$$

$$\begin{aligned}\delta W &= \delta W_1 + \delta W_2 \\ &= -F\delta v + F'\delta v' \\ &= \left(-Fr_F + F\left(r_F + a\right)\right)\delta\varphi \\ &= Fa\delta\varphi \\ &= M\delta\varphi\end{aligned}$$

# 10.3.2. Virtual Work

- Virtual Work of a Line Load




$$dF = q(x) \cdot dx$$
$$d(\delta W) = q(x) \cdot dx \cdot \delta w(x)$$
$$\delta W = \int q(x) \cdot \delta w(x) \cdot dx$$

Computation via Resulting Force:

with $\delta w(x) = \delta w_R + \bar{x} \cdot \delta\varphi$

$$\delta W = \int q(\bar{x}) \cdot (\delta w_R + \bar{x} \cdot \delta\varphi)\, d\bar{x} = R \cdot \delta w_R$$

# 10.3.2. Virtual Work

## Equilibrium Condition with Virtual Work

- After summing up all the actions to $\vec{R}$ and $\vec{M}$ the virtual displacements $\delta\vec{u}$ and virtual rotation $\delta\vec{\varphi}$ perform the virtual work

Box 10.7

- In case of static Equilibrium ($\vec{R} = \mathbf{0}$ and $\vec{M} = \mathbf{0}$) it follows:

Box 10.8

# 10.3.4. Calculation of support and inner reactions with the principle of virtual work

Procedure:

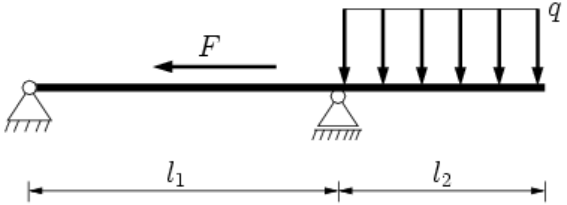
- In order to be able to apply PvW to statically determinate supported systems, the required support reaction is cut free, and the corresponding force effect is applied.

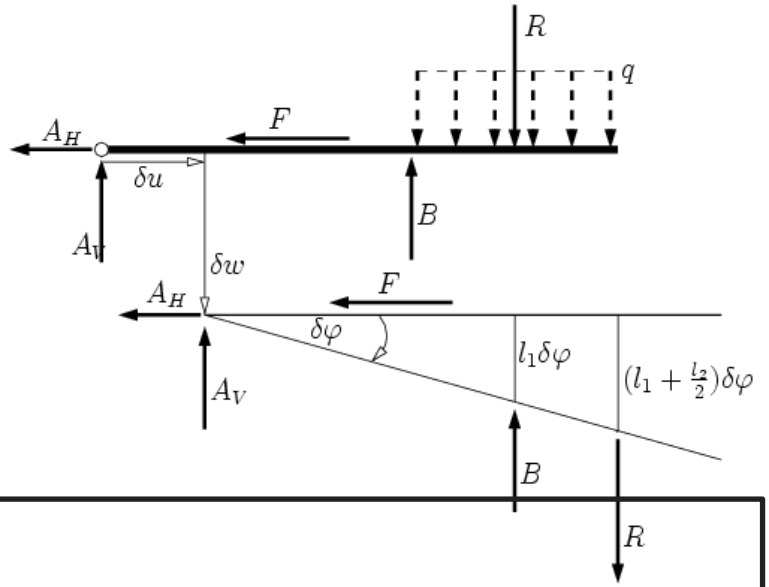
- A displaceable system is created.
- Freed force action performs work.

- The **sum** of all virtual work (signed) in the system must disappear for equilibrium condition.

# 10.3.4. Calculation of support and inner reactions with the principle of virtual work

- Example



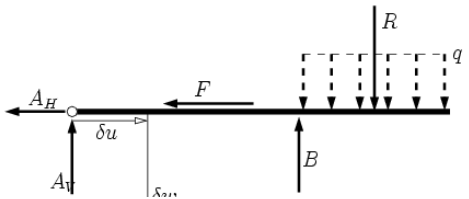




Support Reactions:

Box 10.8

# 10.3.4. Calculation of support and inner reactions with the principle of virtual work

Since $\delta\vec{u}$; $\delta\vec{w}$ and $\delta\vec{\varphi}$ are linearly independent, arbitrary displacements, the total virtual work can only become zero, if in each case the factors in square brackets disappear.




Box 10.10

# 10.3.4. Calculation of support and inner reactions with the principle of virtual work

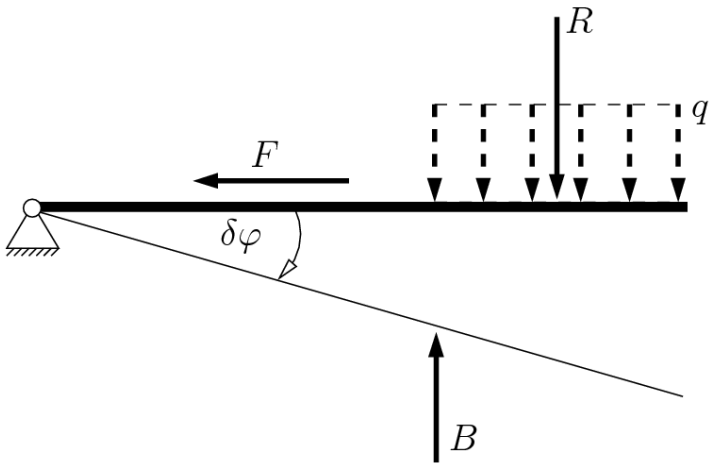
Principle of virtual work leads to the same equations as the previous equilibrium conditions.

But the PvW is more **flexible**:

- If only **one** support reaction is sought, only this reaction is solved!
- Via kinematic relations the displacement figure is obtained.
- Virtual work leads to a single equation for the determination of the wanted force quantity.

# 10.3.4. Calculation of support and inner reactions with the principle of virtual work

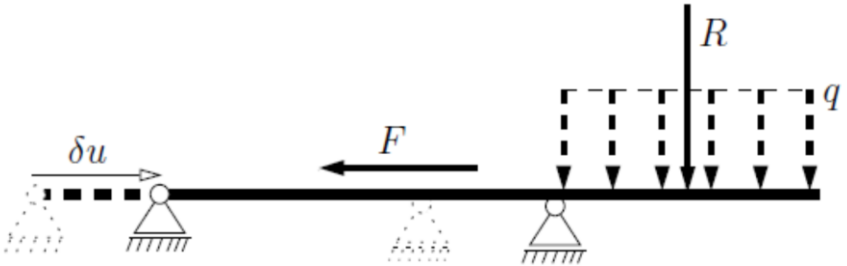
- Wanted: B -> virtual displacement $\delta\varphi$:




Box 10.11

# 10.3.4. Calculation of support and inner reactions with the principle of virtual work

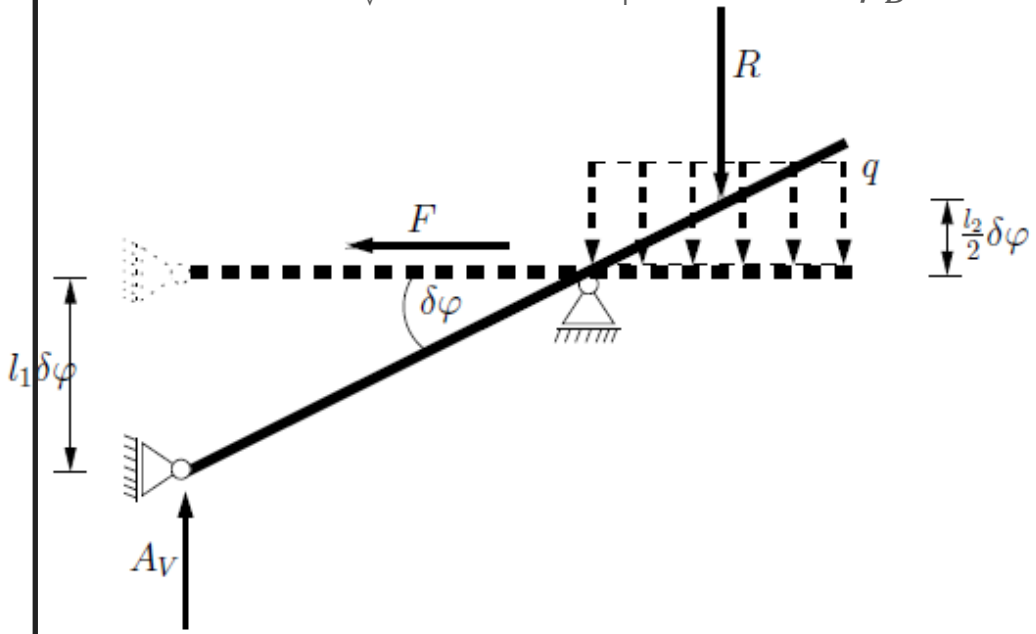
- Wanted: $A_H$ -> virtual displacement $\delta u$:




Box 10.12

# 10.3.4. Calculation of support and inner reactions with the principle of virtual work

- Wanted: $A_V$ -> virtual displacement $\delta\varphi_B$:



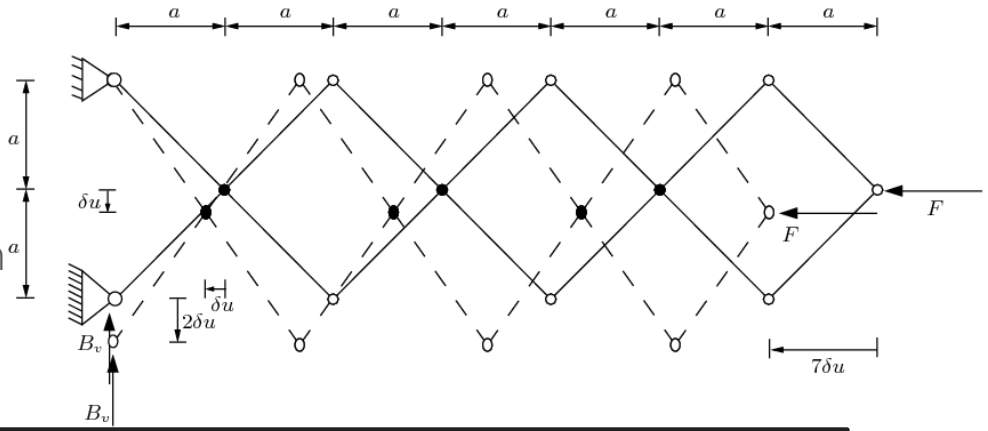

Box 10.12

# 10.3.4. Calculation of support and inner reactions with the principle of virtual work

- Often superior calculation method compared to setting up the equilibrium conditions.

- **Example for clarification:** Scissor lift from hinged connected:

  Vertical support force $B_V$ wanted.




Box 10.14



Autonomous Aerial Systems

## Calculation of inner reactions with the principle of virtual work

Procedure:

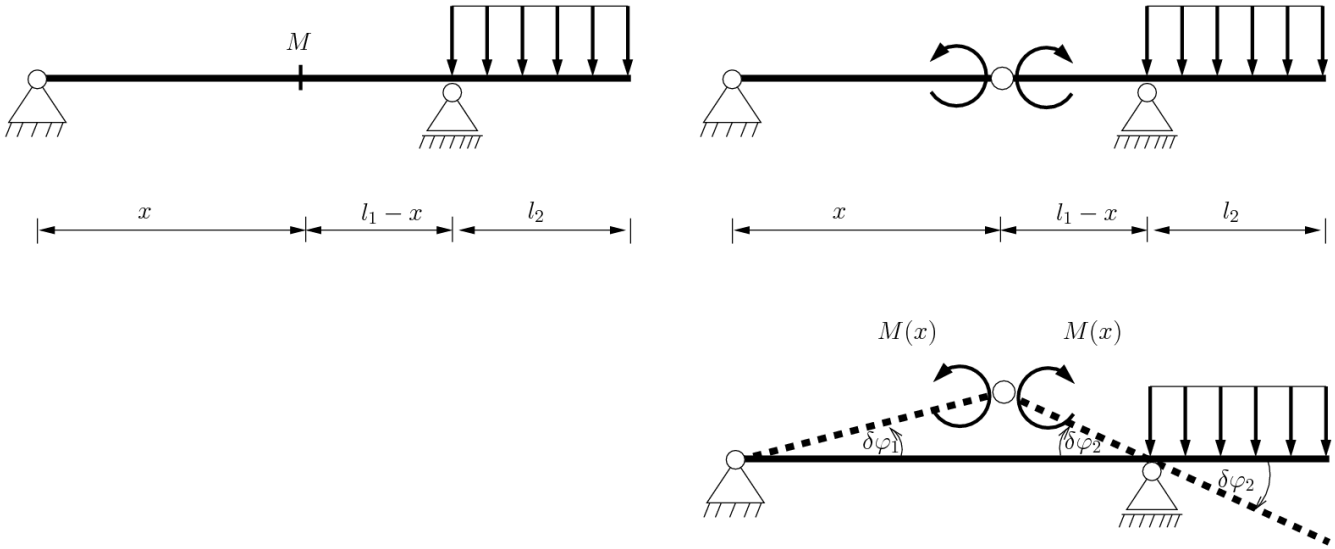
- As before (support reactions), the required section is cut free.
- Again, the kinematic of the complete system is created.
- A virtual displacement is applied.
- The sum of all virtual works must result in 0.

- Specific internal forces can be calculated very easily

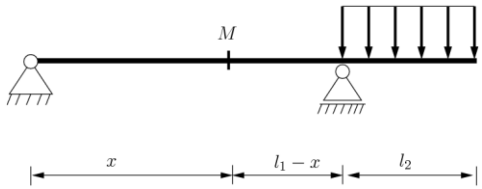
# 10.3.4. Calculation of support and inner reactions with the principle of virtual work

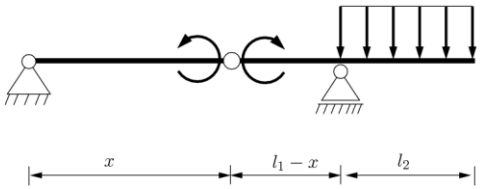
Example: Bending Moment M(x):

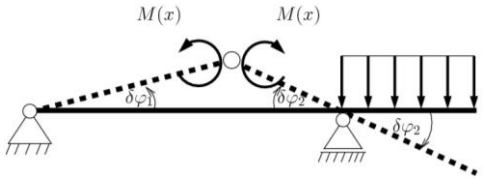
- Free cutting of the moment leads to a rotational joint at this point and the two internal reactions



# 10.3.4. Calculation of support and inner reactions with the principle of virtual work



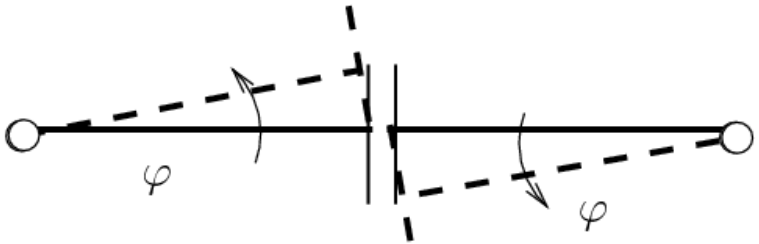




Box 10.14

# 10.3.4. Calculation of support and inner reactions with the principle of virtual work

- Computation of Shear Forces ↦ Shear Force Joint
- Computation of Normal Forces ↦ Normal Forces Joint

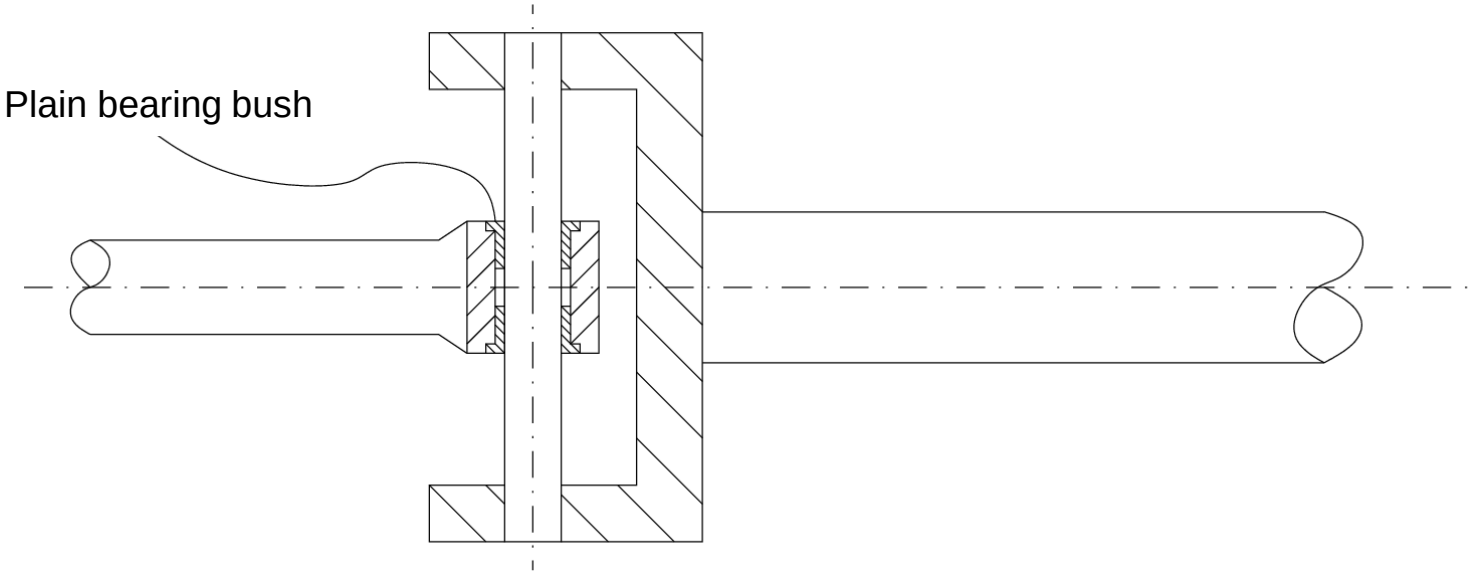



Shear Force Joint

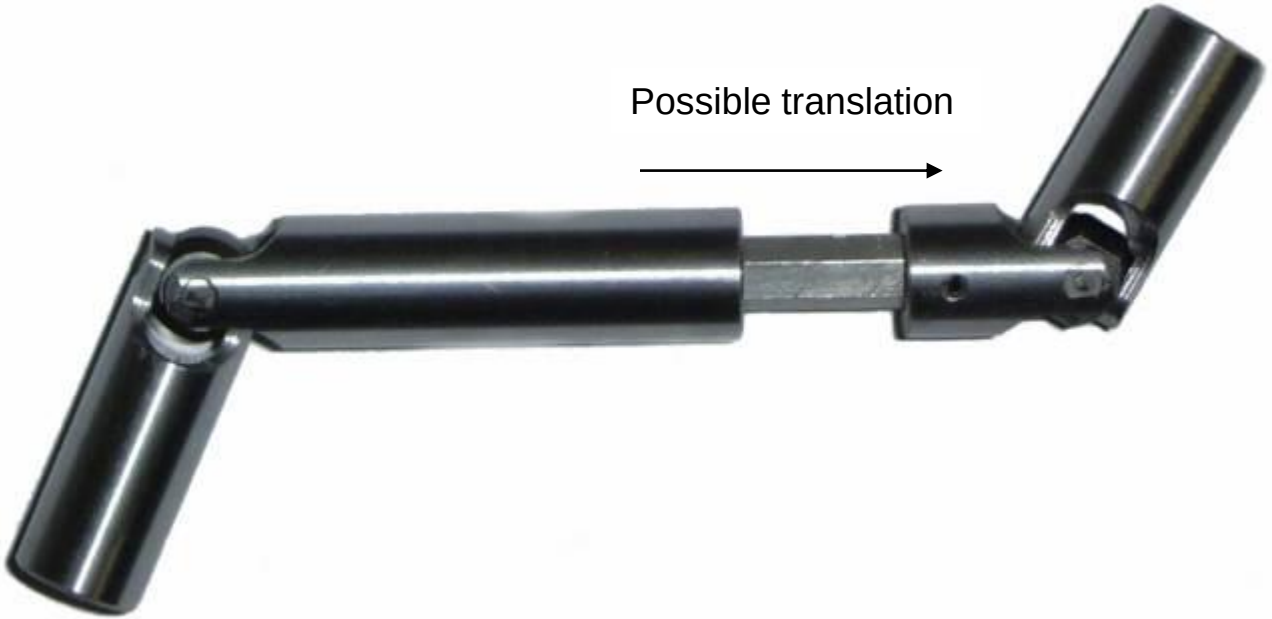


Normal Force Joint

## Construction of a Shear Force Joint

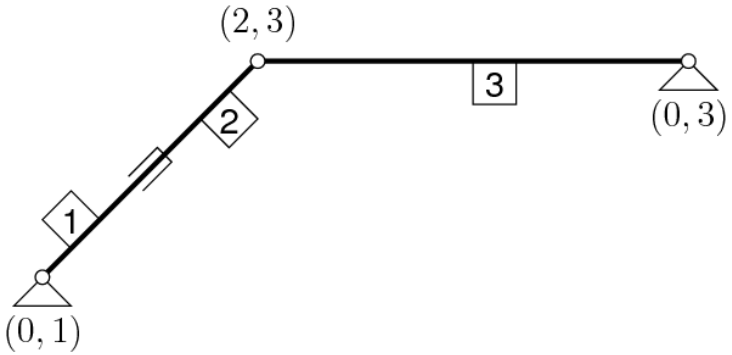


## Normal Force and Moment Joint



# 10.3.4. Calculation of support and inner reactions with the principle of virtual work

## Pole Plan / Normal Force Joint



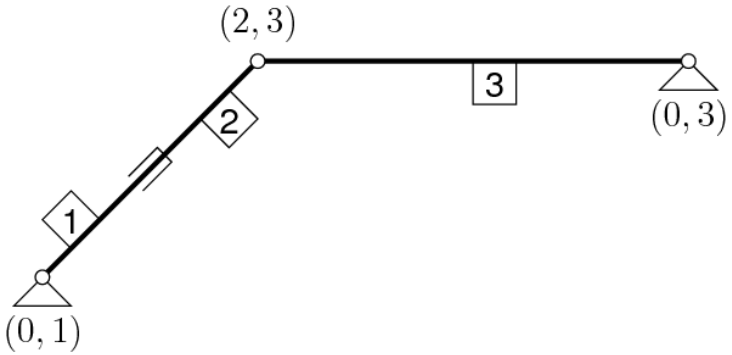

Box 10.16



Autonomous Aerial Systems

# 10.3.4. Calculation of support and inner reactions with the principle of virtual work

## Pole Plan / Shear Force Joint




Box 10.17



Autonomous Aerial Systems